Sreenath Karthick Muthuvel

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SUMMARY

Computer Science undergraduate at IIT Palakkad with experience in programming languages, systems, formal verification, and software engineering. Experienced in compiler construction, formal reasoning with Rocq, systems programming, distributed systems, and software tooling. Interested in applied research at the intersection of programming languages, formal methods, systems, and trustworthy software.

EDUCATION

Indian Institute of Technology Palakkad *Aug 2023 – Apr 2027*
B.Tech in Computer Science and Engineering **CGPA: 9.55/10.0**

TOEFL iBT: 110/120 (**CEFR: C1**)
GRE: 317.5/346 (Quantitative Reasoning 165/170)

EXPERIENCE

Research Intern *May 2025 – Jul 2025*
Defence Research and Development Organisation (DRDO: DYSL-CT), Chennai

- Researched and implemented Explainable AI techniques including **LIME, SHAP, and Class Activation Maps** to interpret deep-learning model predictions.
- Developed **PeliCAM**, a Python/PyQt application using **PyTorch** for model interpretability, supporting CAM analysis, LIME explanations, saliency visualization, and debugging across 3+ neural-network models.
- Contributed to text-classification systems for English and Indian languages.
- GitHub Repository

Software Intern *Feb 2025 – Dec 2025*
Learning Sciences Research Group, Homi Bhabha Centre for Science Education (HBCSE – TIFR)

- Developed interactive web-based tools using **JavaScript, HTML, and CSS** for mathematical learning and experimentation.
- Contributed to a research project analyzing the learning processes of **100+ students**.
- Co-authored a peer-reviewed paper published in the *Proceedings of the 33rd International Conference on Computers in Education orgainesed by IIT Madras (ICCE 2025).*
- GitHub Repository
- Link to Conference Paper

Custom Software Engineering Associate *May 2026 – Jul 2026*
Accenture – Coimbatore

- Gained hands-on experience with Microsoft Azure security, including Virtual Machines, Network Security Groups, RBAC, Azure Policies, and Log Analytics.
- Worked with Azure monitoring and security tools including **Azure Monitor** and **Microsoft Defender for Cloud**.
- Performed security-log analysis using **Kusto Query Language (KQL)**.

PROJECTS

Gaur Tool Chain - Formal Verification *GitLab*

- Developing a software toolkit for the Foundations of Computing Systems course based on a custom **32-bit RISC-V-inspired ISA**, including an assembler, disassembler, CPU simulator, and VM emulator.
- Using **Rocq** to formally reason about the correctness of instruction representations, parsing, assembly, and simulation components.
- Developing system infrastructure for instruction representation, parsing, assembly, simulation, and verification.
- *Tools:* OCaml, Rocq, js_of_ocaml.

Tiger Compiler - Compiler for a Subset of the Tiger Language *GitLab*

- Developed a compiler for a subset of the Tiger programming language in **OCaml**, supporting assignments, arithmetic expressions, `print`, `for`, `if-then`, and `if-then-else` constructs.
- Implemented the compilation pipeline from lexical analysis and parsing through AST construction, Tree IR, canonicalization, basic-block formation, and MIPS code generation.
- Built compiler infrastructure for control-flow analysis, including basic-block construction and graph modules.
- Implemented naive register allocation with spilling and generated executable MIPS assembly compatible with the SPIM simulator.
- *Tools:* OCaml, Menhir, OCamllex, MIPS Assembly, SPIM.

Distributed Rate Limiting System with Redis and Lua *GitHub*

- Implemented Token Bucket, Leaky Bucket, Fixed Window, Sliding Window Log, and Sliding Window Counter algorithms.
- Built atomic **Redis + Lua** implementations to maintain correctness under concurrent and distributed workloads.
- Developed CI-based benchmarks to compare algorithmic performance under simulated traffic patterns.
- *Tools:* Python, Redis, Lua, GitHub Actions.

SageServe - MLOps for LLM Serving Optimization *GitHub*

- Developed an **MLOps pipeline** around SageServe for workload forecasting and simulation-based optimization of LLM serving infrastructure.
- Implemented data processing, validation, forecasting, model evaluation, and simulation workflows following reproducible MLOps principles.
- Integrated **Docker** for reproducible environments and **GitHub Actions** for automated CI workflows.
- *Tools:* Python, Docker, GitHub Actions, Pandas, NumPy, Scikit-Learn, Git.

COP - Police Workforce Incident Management System *GitHub*

- Developed an incident management system for police workforce operations as part of a team-based software engineering project.
- Implemented application functionality using **C#** and **WinUI**, following software engineering practices for collaborative development.
- Used **Git and GitHub** for version control, collaboration, and project management.
- *Tools:* C#, WinUI, Git, GitHub.

Local LLM Debate System for Hallucination Reduction *GitHub*

- Designed a local multi-agent LLM framework using **LangGraph**, where independent agents answer, critique, revise, and evaluate responses through multiple stages.
- Built a resource-routing pipeline that dynamically selects between **web search, RAG, local document retrieval, and Python tools** for evidence-grounded reasoning.
- Developed an event-driven architecture with a thread-safe event bus and real-time monitoring of routing decisions, retrieved evidence, debate rounds, and model outputs.
- *Tools:* Python, LangGraph, LangChain, Ollama, Qwen, ChromaDB, HuggingFace Embeddings, Rich, Pandas.

Sheet2Slide - Agentic Workflow *GitHub*

- Built a multi-stage agentic AI pipeline using **LangGraph** that transforms structured spreadsheet data into presentation-ready reports.
- Engineered structured-output LLM agents for analytical reasoning while delegating statistical computation and visualization to deterministic Python tools.
- *Tools:* Python, LangGraph, Ollama, LangChain, Pandas, Matplotlib, Python-PPTX, Pydantic.

TECHNICAL SKILLS

- **Formal Methods & PL:** Rocq, Coq/SSReflect, Formal Verification, Proof Assistants, Dependent Types, Compiler Construction, Type Systems
- **Systems & Programming:** OCaml, C, C++, Python, C#, MIPS Assembly, JavaScript, SQL
- **Systems Engineering:** Linux, Redis, Lua, REST APIs, Distributed Systems, Git, GitHub Actions
- **Compiler Technologies:** Menhir, OCamllex, ASTs, Tree IR, Control-Flow Graphs, Basic Blocks, Register Allocation, Code Generation
- **Software Engineering:** Git, GitHub, Testing, CI/CD, Docker, WinUI
- **AI / ML:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, LangGraph, LangChain, Ollama, SHAP, LIME, CAM, Pandas, NumPy, Matplotlib
- **Web & Databases:** FastAPI, Flask, React, PostgreSQL, HTML, CSS

ACHIEVEMENTS & LEADERSHIP

- **Inter-IIT Tech Meet 13.0:** Secured **8th position nationally** in the ISRO Problem Statement track.
- **Subsystem Lead – Avionics, Rocketry Team:** Led the avionics subsystem for the INSPACE Model Rocketry competition, coordinating hardware/software integration and subsystem tasks. **Secured a top-15 position nationally**.
- **Core Team Member, NSS IIT Palakkad:** Coordinated and executed social service events on campus.